

ANDREW CRONIN

+50766541186 | andrewcronin08@gmail.com
154B Williamson Avenue | Gamboa, Colón, Panamá

Curriculum Vitae

RESEARCH SYNOPSIS:

My research interests combine animal behavior, sexual and natural selection, and phenotypic plasticity. Previous work has largely focused on the implications and effects of anthropogenic changes (particularly urbanization and climate change). I enjoy merging observational and experimental research in both field and lab settings. Though I am interested in diverse taxonomic groups, my research has primarily focused on amphibians.

EDUCATION:

- Oct 2024 *Ph.D., Vrije Universiteit Amsterdam*, The Netherlands (with Dr. W. Halfwerk & Dr. J. Ellers)
Thesis title: *Calling males and tadpole tales: Effects of urbanization across different life stages in the túngara frog*
- May 2018 *M.S. in Applied Biology, Salisbury University*, Salisbury, MD (with Dr. R. Taylor & Dr. K. Hunter)
Thesis title: *Environmental heterogeneity influences female mate choice in the túngara frog*
- May 2012 *B.A. in Biology, Connecticut College*, New London, CT
Minor in East Asian Studies (Chinese Focus)

RESEARCH EXPERIENCE

- 2025 – Present *NSF and STRI Postdoctoral Fellow, University of North Carolina at Chapel Hill*, Chapel Hill, NC and *Smithsonian Tropical Research Institute*, Panamá
- Conduct experiments and data analysis on social dynamics in response to predation risk
 - Experimentally examine the specificity of phenotypic plasticity in *Engystomops pustulosus*
 - Examine relationships between parental traits and offspring plasticity in *Engystomops pustulosus* and *Spea multiplicata*
 - Examine long-term carry over effects of larval polymorphisms in *Spea multiplicata*
- 2019 – 2024 *PhD Candidate, Vrije Universiteit Amsterdam*, The Netherlands
- Designed and conducted field and lab experiments investigating environmental and evolutionary effects of urbanization on tadpole development and behavior
 - Coordinated and conducted field experiments examining direct and indirect effects of light and noise pollution of male calling behavior in *E. pustulosus*
 - Quantified selection pressures on male *E. pustulosus* in response to light and noise pollution
 - Developed and maintained common garden breeding experiment
- 2015-2018 *Graduate Student, Salisbury University*, Salisbury MD
- Examined *E. pustulosus* responses to multimodal stimuli, audio-visual grouping, spatial ecology, and temporal changes in female mate choice
 - Designed and conducted phonotaxis experiments
 - Analyzed call traits in *Hyla cinerea* using SoundRuler, Audition, and Raven
 - Assisted in developing study examining spatial temporal patterns of *E. pustulosus* via radiotelemetry
 - Prepared reduced representation genomic libraries and analyzed genome wide single nucleotide polymorphisms
- 2014-2015 *Research Technician, University of Missouri*, Balcones Canyonlands N.W.R., Lago Vista TX
- Mapped territories and determined nesting success in the endangered golden-cheeked warbler
 - Managed prescribed burn vegetation surveys
- 2013 *Research Technician, University of Illinois*, Savannah River Ecology Laboratory, Aiken SC
- Tracked gray rat snakes and black racers using radiotelemetry

- Monitored bird nests and analyzed video for behavioral data of birds and of predators
- Collected vegetation data and estimated small mammal abundance

2012 *Biological Field Technician, Arizona State University*, Sevilleta N.W.R., La Joya NM

- Collected behavioral, hormonal, and spatial data on lizards in a lab and outdoor arenas
- Set up and maintained field as well as lab arenas examining thermal preferences

2010 *Research Fellow, Jug Bay Wetlands Sanctuary*, Lothian MD

- Monitored box turtle populations and tracked mud turtles using radiotelemetry
- Designed research project examining the basking behavior of northern water snakes
- Presented findings in a research paper and in a presentation to the public

TEACHING AND SUPERVISING EXPERIENCE

April 2025 *Field Course Lecturer and Mentor, McGill University*, Canada

- Joint led *Sensory Ecology* module for McGill-Neo graduate school program
- Presented lectures on sensory pollution and mentored student research projects

2019-2024 *PhD Candidate, Vrije Universiteit Amsterdam*, The Netherlands

- Supervised three MSc students and two BSc students, and co-supervised two MSc students
- Guest lecturer for MSc course *Metropole Ecology* and bachelor's *Evolutionary Developmental Biology* course
- Developed and implemented practical examining human sexual selection for bachelor's *Evolutionary Developmental Biology* course
- Led discussion groups and evaluated presentations for bachelor's course *Adaptation to the Human Environment*

Summer 2023, 2024 *Course Coordinator, Vrije Universiteit Amsterdam*, The Netherlands

- Coordinated and led second year bachelor's course *Urban Ecology*
- Designed curriculum based on constructive alignment principles and problem-based learning

Feb-June 2023 *University Teaching Qualification Program, Vrije Universiteit Amsterdam*, The Netherlands

- Demonstrated ability to design curriculum and learning objectives
- Developed learning activities and supervision plans utilizing educational and motivation theory
- Created portfolio of my teaching and supervision practices, culminating in certification

2018-2018 *Teaching Fellow, Barrie School*, Silver Spring MD

- Developed curriculum and was lead instructor for *Life Science* (7th grade course with 25 students)
- Head coach of the middle school soccer team

2015-2018 *Teaching Assistant, Salisbury University*, Salisbury MD

- Led *Anatomy and Physiology* lab for six semesters, teaching 70 students each semester
- Led guest lectures for upper-level bachelor's *Genetics Analysis* course and assisted in training next generation genetic sequencing techniques
- Trained three bachelor's students in experimental and genetics methods in Gamboa, Panamá

VOLUNTEER EXPERIENCE

2021-2024 *Volunteer, Reptielen Amfibieën Vissen Onderzoek Nederland*, Amsterdam, Netherlands

- Monitored grass snake and sand lizard populations with transect checks
- Collected DNA samples to examine hybridization of *Pelophylax* frog species

2014-2015 *Volunteer, University of Texas*, Austin, TX

- Assisted Dr. Sofia Rodriguez-Brenes with experimental design, and maintained túngara frog colony (Dr. M. Ryan lab)
- Assisted Dr. Brett Pasch, collecting singing mice and conducting transmission experiments in the field

(Dr. S. Phelps lab)

- 2013-2014 *Volunteer, Chelonia Ecological Adventures*, Severna Park, MD
- Educated children about reptiles and reptile conservation
 - Provided animal care, including cleaning, feeding, handling, and monitoring
- 2011 *Volunteer, Zanzibar Zoological Animal Park*, Zanzibar, Tanzania
- Assisted with tours and educated visitors about local reptile diversity and conservation efforts
 - Assisted park director and staff with maintenance and behavioral monitoring
- 2011 *Researcher, Jozani National Park*, Zanzibar, Tanzania
- Designed study examining lizard populations in agricultural and forest habitats
 - Analyzed data and presented findings in a report and oral presentation

GRANTS AND FELLOWSHIPS AWARDED

- 2025-Present National Science Foundation Postdoctoral Research Fellowship in Biology (PRFB) (Advisors: Dr. David Pfennig & Dr. Karin Pfennig)
- 2025-Present Smithsonian Tropical Research Institute (STRI) postdoctoral fellowship (Advisor: Dr. Rachel Page)
- 2021-2022 Smithsonian Tropical Research Institute (STRI) predoctoral fellowship (Advisor: Dr. Rachel Page)
- 2016 Salisbury University Research and Presentation Grant
- 2015 Salisbury University Research and Presentation Grant
- 2015 Salisbury University Henson School Graduate Research Fund
- 2010 Jug Bay Undergraduate Research Fellow

PUBLICATIONS (PUBLISHED)

***Cronin, A.D.**, Zilber R., Jerem, P.M., Halfwerk, W. 2025. Noise pollution and artificial light at night alter selection pressures on sexual signals in an urban adapter. *Journal of Evolutionary Biology*, 38(10): 1410–1420.

<https://doi.org/10.1093/jeb/voaf092>

*Selected as Editor's Choice

Cronin, A.D., Smit, J.A.H., Ellers J., Halfwerk, W. 2025. Urbanization drives intra- and intergenerational developmental phenotypes in the túngara frog, *Engystomops pustulosus*. *Journal of Animal Ecology*, 94: 1707–1719.

<https://doi.org/10.1111/1365-2656.70071>

Cronin, A.D., Taylor, R., Page, R., Ryan, M., Murphy, M., Hunter, K. 2025. Rapid cross-generational changes in morphology, genetics, and mate choice following an extreme climatic event. *Evolutionary Ecology*, 39: 495–508.

<https://doi.org/10.1007/s10682-024-10324-1>

Smit, J.A.H., Thijssen, V., **Cronin, A.D.**, Ellers J., Halfwerk, W. 2024. Urban sensory conditions alter rival interactions and mate choice in urban and forest frogs. *Behavioral Ecology*, 35(6), arae088. <https://doi.org/10.1093/beheco/arae088>

Cronin, A.D., Smit, J.A.H., and Halfwerk, W. 2022. Anthropogenic noise and light alter temporal but not spatial breeding behavior in a wild frog. *Behavioral Ecology*, 33(6): 1115-1122. <https://doi.org/10.1093/beheco/arac077>

Smit J.A.H., **Cronin A.D.**, van der Wiel I., Oteman B., Ellers J., Halfwerk W. 2022. Interactive and independent effects of light and noise pollution on sexual signaling in frogs. *Frontiers in Ecology and Evolution*, 10: 934661.

<https://doi.org/10.3389/fevo.2022.934661>

Cronin, A.D.¹, Smit, J.A.H.¹, Muñoz, M.I.¹, Poirier, A., Moran, P.A.¹, Jerem, P.¹ and Halfwerk, W.¹. 2022. A comprehensive overview of the effects of urbanisation on sexual selection and sexual traits. *Biological Reviews*, 97(4): 1325-1345.

Cronin, A.D., Ryan, M., Page, R., Hunter, K., Taylor, R. 2019. Environmental heterogeneity alters mate choice behavior for multimodal signals. *Behavioral Ecology and Sociobiology* 73(43): 1–10. <https://doi.org/10.1007/s00265-019-2654-3>

Garcia, M., **Cronin, A.D.**, Bowling, T., Bushera, H., Hunter, K., Taylor, R. 2019. Dueling Frogs: Do male green treefrogs (*Hyla cinerea*) alter their calling behaviors in response to surrounding competitor call frequency? *Behavioral Ecology and Sociobiology* 73(21): 1–10. <https://doi.org/10.1007/s00265-018-2632-1>

PUBLICATIONS (IN REVIEW)

Smit J.A.H., **Cronin A.D.**, Moran, P.A., Ellers J., Halfwerk, W. *In review*. Heritable differences in sexual signaling and morphology between urban and forest túngara frogs.

PRESENTATIONS

- | | |
|------|--|
| 2023 | Dutch Behavioral Society Meeting. Egmond aan zee, Noord-Holland, The Netherlands.
Oral Presentation: “Urbanization as a driver of phenotypic changes across multiple life stages”. |
| 2023 | Society of Integrative and Comparative Biology Meeting, Austin, TX, USA.
Oral Presentation: “Effects of urbanization on early life stages and potential for local adaptation” |
| 2022 | Dutch Behavioral Society Meeting. Egmond aan zee, Noord-Holland, The Netherlands.
Oral Presentation: “Effects of urbanization on early life stages and potential for local adaptation” |
| 2021 | Dutch Behavioral Society Meeting. Egmond aan zee, Noord-Holland, The Netherlands.
Oral Presentation: “Direct and Indirect Influences of Multisensory Pollution on Sexual Communication” |
| 2020 | Animal Behavior Society Conference, Online.
Video Presentation: “Direct and Indirect Influences of Multisensory Pollution on Sexual Communication” |
| 2020 | Netherlands Annual Ecology Meeting, Bunnik, Utrecht, The Netherlands.
Oral Presentation: “Predators, Parasites, and Pale Lights” *awarded best presentation |
| 2018 | Animal Behavior Society Conference, Milwaukee, WI, USA.
Oral Presentation: “The frogs, they are a changin’: Extreme climatic events alter mate choice behavior” |

OUTREACH

- | | |
|-----------|--|
| 2026 | Noche de Bichos (Smithsonian Tropical Research Insitute and Biomuseo, Panamá City, Panamá) |
| 2025 | Invited blog post: Animal Ecology in Focus “Life in the Fast Lane: Developmental Responses of Tadpoles to Life in the City”
https://animalecologyinfocus.com/2025/07/01/life-in-the-fast-lane-developmental-responses-of-tadpoles-to-life-in-the-city/ |
| 2022-2026 | Bat Night Presentation (Smithsonian Tropical Research Institute, Gamboa, Panamá) |
| 2022 | Invited blog post: Smithsonian Magazine “Life in the (Urban) Jungle”
https://www.smithsonianmag.com/blogs/smithsonian-institution-office-fellowships-and-int/2023/03/06/life-in-the-urban-jungle/ |

PROFESSIONAL SERVICES

- Organization of the Urban Ecology symposium “Evolving Urban Communities” at Vrije Universiteit Amsterdam
- Reviewer for the following journals: *Animal Behavior*, *Animal Cognition*, *Behavioral Ecology and Sociobiology*,

